MATÍAS FERNÁNDEZ LAKATOS

PhD in Optical Physics & Msc in Theoretical Physics & MSc in Big Data

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CV UY ANII

PROFESSIONAL EXPERIENCE

Research Engineer

2025 – present — Gradiant, Vigo, Spain

- Development of intelligent solutions for early threat detection in corporate cybersecurity.
- Design and implementation of unsupervised machine learning models (β-Variational Autoencoder, Autoencoder, Extended and Isolation Forest) for real-time anomaly detection and behavior analysis in large-scale data environments.
- Focus on model explainability (XAI) to enhance usability for non-technical users.
- Integration into a scalable data architecture using technologies such as Kafka, Spark, and Airflow for efficient real-time data processing.

Researcher and Teaching Assistant at UdelaR

2015 – 2024 — Fac. Ingeniería, Udelar, Montevideo

- Taught university physics at the Physics Institute.
- Conducted research in multiple projects, including study of long-range properties of the strong nuclear interaction, development of multispectral instruments for remote monitoring, phase object retrieval involving a new method and creating a new method of integration from just one derivative.

UNIVERSITY DEGREES

MSc Tecnologías de Análisis Masivo de Datos: Big Data

2024 – 2025 — USC, Santiago de Compostela

- Thesis: Application of new anomaly detection models in Big Data contexts. Courses on Large-Scale Databases, Technologies for Managing Unstructured Information, Computing Technologies for Big Data, IoT, Statistics, Data Mining, Data Visualization, Business Intelligence, and Applications and Use Cases in Business.

PhD in Optics

2019 – 2024 — Udelar, Montevideo

- Thesis: Visualization and Characterization of Phase Objects (clickable) Courses on Laboratory of fundamental electronics and scientific instrumentation, Coherent Optics, **Reinforcement Learning**, **Computer Image Processing**, **Computational Multivariate Statistics** and **Deep Learning for Computer Vision (cs231n)** . Advisors: PhD. José A. Ferrari, PhD Erna Frins and PhD Gastón A.Ayubi.

MSc in Quantum Chromodynamics

2016 – 2018 — Udelar, Montevideo

- Thesis: Rol de los diversos acoplamientos en la cromodinámica cuántica infrarroja (clickable) Courses on Statistical Mechanics, Quantum Field Theory I and II, General Relativity. Advisors: PhD. Nicolás Wschebor and PhD Marcela Peláez.

Bachelor's Degree in Physics

2011 – 2015 — Udelar, Montevideo

ABOUT ME

I'm a research engineer with a background in Physics (Ph.D. and MSc) and Big Data Analytics (MSc.) My work focuses on intelligent systems for anomaly detection, large-scale data processing, and applied machine learning. I've presented my research at conferences and events, refining my scientific communication skills. With over nine years of teaching and community service — including housing projects and high school classes — I've developed strong teamwork, empathy, and leadership abilities, blending technical depth with human-centered collaboration. Outside of work, I enjoy board games and reading, embracing the idea of "journey before destination" as a mindset for continuous learning and creative exploration.

ACHIEVEMENTS

Master's scholarship from the Comisión Académica de Posgrados (CAP, 2016)

PhD scholarship from the Agencia Nacional de Investigación e Inovación (ANII,2019)

PhD scholarship from the Comisión Académica de Posgrados (CAP,2022)

Member of the National System of Researchers in Uruguay (SNI, 2024).

4 first author peer-reviewed articles.

clickable, 2019, IJMPA Scopus
clickable, 2022, Optik Scopus
clickable, 2023, OLE Scopus
clickable, 2024, SPIE

Author peer-reviewed article

clickable, 2024, IOP Science
clickable, 2025, OLT

TECHNICAL SKILLS

Git | Data analysis | R | SQL | Python | Machine Learning | modeling | Matlab | Digital Image Processing | Linux | Statistical Analysis | LaTeX | Multivariate Analysis | Optimization | Database Management and Modeling